

Homework 9

Due: Thursday 5/22 at 11:59pm

Complete the following problems and submit through Google Classroom. For the first three problems, you can submit a picture of paper solutions, a pdf, Google Doc, etc. For the programming problem, submit a .zip file. Name all your files based on your name or username.

1. (8 points) Consider the array of integers

4	3	7	1	5	6	2
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- (a) Draw the sequence of swaps (following the format we saw on 4/23 in lecture) that result from sorting this array using selection sort.
- (b) Repeat, but using heapsort (as described in class) as the sorting algorithm.
2. (8 points) The following array contains a min-heap (a heap where each node is smaller than its children):

1	5	2	6	10	4	9	7	8
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- (a) Draw this heap in tree format.
- (b) Show the heap that results if you add 3 to it. Display the heap in both the array and tree formats.
- (c) Show the heap that results if you dequeue from the original heap (NOT the version from part (b)). Display the heap in both the array and tree formats.
3. (7 points) Use radix sort (by digit) to sort the following numbers:

104, 213, 300, 411, 20, 102, 244, 134, 4, 200, 303, 302, 214, 1, 13, 402, 100, 231.

(For convenience, each digit is 0–4; you only have to include buckets for these digits.) Show the bucket contents and resulting list after each digit is considered. (So I want 3 figures and lists...) The numbers should start in the order they are given above. Note that you do not need to implement radix sort with a program, though you are welcome to do so.

4. (8 points) Modify the program `SumArray` from Lab 8 so that it sums the values in a 100×100 matrix (2D array). The base case (threshold for resorting to serial computation) should be a single row or column of the matrix.
5. (1 point) At the top of the file containing your solution for problem 1, include a comment with a statement about your collaboration (if any).